Robust Speech Accent Recognition: A Comparative Study of Classical and Self-Supervised Learning Architectures

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Abstract

Accent Recognition (AR) is a fundamental component for enhancing the inclusivity and robustness of Automatic Speech Recognition (ASR) systems, yet it remains a significant challenge due to the high variability of non-native speech and limited labeled data [1]. Current ASR models often suffer severe performance degradation when encountering diverse accents [2], and classical classification methods frequently lack the capacity to generalize effectively. In this paper, to address these limitations, we present a comprehensive comparative analysis of classical machine learning baselines (Random Forest [3], SVM [4], XGBoost [5], k-NN [6]) versus state-of-the-art Self-Supervised Learning (SSL) architectures, including **wav2vec2-base** [7], XLSR-53 [8], and **w2v-bert-2.0** [9]. Utilizing the Speech Accent Archive [10], we introduce a noise-augmented fine-tuning strategy to improve model robustness against acoustic variability. Experiments demonstrate the significant superiority of SSL frameworks: while classical models plateaued at **29.44%** accuracy, our fine-tuned **w2v-bert-2.0** model achieved a state-of-the-art accuracy of **52.1%,** surpassing both the baseline and other SSL variants. This highlights the effectiveness of leveraging large-scale pre-trained transformer representations for low-resource accent classification tasks.

1. Introduction

The ubiquity of voice-activated interfaces, ranging from smartphones to smart home environments, has established Automatic Speech Recognition (ASR) as a cornerstone of modern human-computer interaction. While these systems have achieved near-human performance on standard dialects, they frequently exhibit significant performance degradation when encountering non-native or regional accents. [2] An accent, manifesting as distinctive phonological and prosodic variations associated with a speaker's locality or social identity, introduces complex acoustic variability that standard ASR models often misinterpret as noise or error. [1] Consequently, Accent Recognition (AR) has emerged not merely as a classification task, but as a fundamental prerequisite for building equitable, adaptive AI systems capable of serving diverse linguistic demographics.

However, despite the critical need for robust AR, the development of reliable models is hindered by the "low-resource" nature of many accents and the high intra-class variability of speech. Classical machine learning approaches, which rely on handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCCs) [11] and traditional classifiers like Support Vector Machines (SVMs), [4] historically hit a performance plateau, failing to capture the nuanced temporal abstractions required to differentiate subtle accent variations. Conversely, fully supervised deep learning architectures typically require vast quantities of labeled data, which is often unavailable for underrepresented accents.

To address these challenges and bridge the gap between theoretical acoustic modeling and real-world robustness, we present a comprehensive comparative study of classical versus self-supervised learning (SSL) architectures for accent classification. Our approach leverages the linguistically diverse Speech Accent Archive [10] to benchmark traditional classifiers against state-of-the-art pre-trained transformer models, including **wav2vec 2.0**, [7] **XLSR-53**, [8] and **w2v-bert-2.0**. [9] Unlike previous approaches that rely on static feature engineering, our methodology employs Transfer Learning [12] to exploit the "universal speech grammar" learned by these massive models, enhancing them via a noise-augmented fine-tuning strategy to improve generalization against environmental perturbations. [13]

The main contributions of our work can be summarized as follows:

1. **Rigorous Benchmarking:** We establish a solid baseline by exhaustively evaluating classical ML algorithms (Random Forest [3] , SVM [4] , XGBoost [5] , k-NN [6]) using MFCC features, demonstrating their limitations with a maximum accuracy plateau of **29.4%**.
2. **SSL Architecture Analysis:** We systematically evaluate and fine-tune three distinct SSL architectures - wav2vec2-base [7]wav2vec2-large-xlsr-53, [8] andw2v-bert-2.0,[9] investigating the impact of model size and pre-training diversity on downstream accent recognition tasks.
3. **Noise-Augmented Robustness:** We introduce a training regimen incorporating white noise injection, [13] which proved critical for generalization. Our best-performing model, **w2v-bert-2.0**, achieved a state-of-the-art accuracy of **52.1%**, significantly outperforming both classical baselines and other SSL variants.
4. **End-to-End Pipeline:** We developed a functional inference pipeline capable of processing raw video inputs into accent predictions, bridging the gap between experimental deep learning and practical application.

2. Related Work

The domain of Accent Recognition (AR) has evolved in parallel with advancements in digital signal processing and machine learning. Early research predominantly focused on feature engineering, while recent paradigms have shifted toward representation learning via deep neural networks.

2.1 Classical Approaches and Feature Engineering

Historically, speech processing systems relied heavily on handcrafted acoustic features to capture the spectral properties of the human voice. The Mel-Frequency Cepstral Coefficient (MFCC) emerged as the de facto standard for feature extraction [11], designed to approximate the nonlinear frequency resolution of the human auditory system by mapping the power spectrum onto the Mel scale [14]. Early accent classification frameworks typically coupled these static features with robust statistical classifiers. Support Vector Machines (SVMs) [4], utilizing kernel functions to map inputs into high-dimensional spaces, were widely adopted for their efficacy in small-data regimes. Similarly, ensemble methods like Random Forest (RF) [3] and gradient boosting frameworks such as XGBoost [5] became popular for their interpretability and resistance to overfitting on structured acoustic data. However, these classical pipelines are fundamentally limited by the information loss inherent in the feature extraction process (e.g., the DCT step in MFCCs decorrelates data but discards fine-grained temporal dynamics), often resulting in a performance plateau when distinguishing between phonologically similar accents.

2.2 The Deep Learning Shift

The introduction of Deep Learning enabled models to learn optimal feature representations directly from raw waveforms, bypassing the need for manual feature engineering. Initially, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were the standard for sequence modeling [15]. However, the advent of the Transformer architecture revolutionized the field [16]. By replacing sequential processing with the Self-Attention Mechanism [16], Transformers can capture long-range prosodic dependencies by relating intonation at the end of an utterance to stress patterns at the beginning, which are critical for identifying dialectal nuances.

2.3 Self-Supervised Learning (SSL) Paradigms

A major bottleneck in AR is the scarcity of labeled accent data. Self-Supervised Learning (SSL) addresses this by pre-training models on vast amounts of unlabeled speech. Wav2Vec 2.0 [7] introduced a framework where a Convolutional Neural Network (CNN) encodes raw audio into latent representations, which are then discretized and processed by a Transformer using a contrastive loss objective. Building on this, XLSR (Cross-Lingual Speech Representations) [8] extended the architecture by pre-training on 53 languages, hypothesizing that exposure to diverse phonological systems enhances adaptability to non-native accents. Most recently, w2v-bert-2.0 [9] combines contrastive learning with Masked Language Modeling (MLM) [17], refining quantization to capture richer contextual information. Our work investigates the efficacy of these SSL architectures against classical baselines, specifically examining their robustness when fine-tuned with noise augmentation [13].

3. Methodology

In this section, we delineate our comprehensive experimental framework, which transitions from traditional feature engineering to advanced self-supervised representation learning. Our approach is bifurcated into two distinct phases: establishing a rigorous baseline using classical machine learning algorithms and subsequently pushing the state-of-the-art with noise-augmented Deep Learning architectures. The high-level architecture of our proposed framework is illustrated in **Figure 1**.

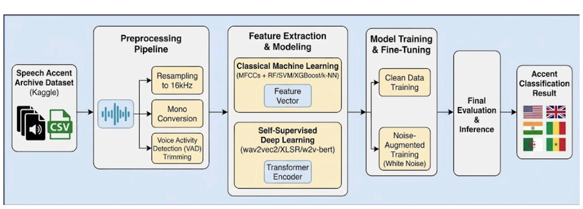

Figure 1: The proposed accent recognition framework. The pipeline begins with the Speech Accent Archive dataset, passes through a rigorous preprocessing stage, and then branches into parallel experimental paths for classical machine learning (using MFCCs) and self-supervised deep learning. The deep learning path is further enhanced by a noise-augmented fine-tuning strategy before final evaluation.

3.1 Dataset and Preprocessing Pipeline

The foundation of this study is the Kaggle Speech Accent Archive, a linguistically rich dataset chosen for its unique control over lexical variability. Every speaker in the archive reads the identical phonetically balanced text, "**The Stellar Williams**" passage. This experimental control is critical; it ensures that acoustic variations in the training data are attributable strictly to speaker accent rather than phonological context.

The metadata, managed via speakers_all.csv, provides the ground truth labels (*country*, *native_language*) while introducing demographic variables (*age*, *sex*) that the model must learn to disregard. To prepare this data for algorithmic consumption, we engineered a unified preprocessing pipeline deployed in our main.py inference engine. As detailed in **Figure 2**, this pipeline consists of four critical stages:

1. **Sanitization:** The dataset manifest was first filtered using the file_missing? flag to purge corrupted or incomplete entries.
2. **Standardization:** All MP3 recordings were transcoded to WAV format and resampled to **16kHz**. This sample rate is non-negotiable as it aligns with the pre-training resolution of the Wav2Vec2 and XLSR backbones.
3. **Channel Reduction:** Stereo signals were downmixed to mono to reduce computational dimensionality.
4. **Voice Activity Detection (VAD):** We applied VAD algorithms to aggressively trim silence from the start and end of utterances, then added a synthesized white noise signal ensuring that our models train solely on vocalized speech segments and background noise.

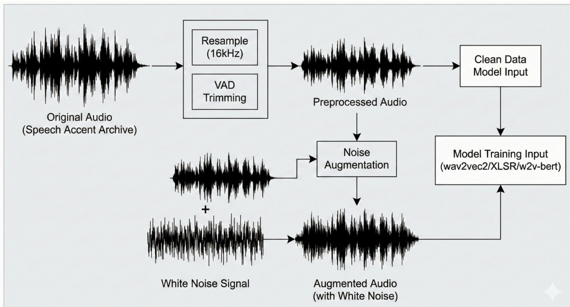

Figure 2: Detailed audio preprocessing and augmentation pipeline. The original audio is resampled to 16kHz and trimmed using VAD. The clean, preprocessed audio is then either fed directly to the model or augmented by adding a synthesized white noise signal, creating a robust training sample.

3.2 Phase 1: Establishing the Classical Baseline

To define the performance floor of current technology, we first explored the limits of static feature engineering. Using the librosa library, we extracted Mel-Frequency Cepstral Coefficients (MFCCs) from the processed audio. Specifically, we computed the mean and variance of the coefficients across time frames to produce a fixed-length feature vector representing the spectral envelope of each speaker.

These feature vectors were fed into four distinct classifiers to evaluate the efficacy of statistical learning:

- **Random Forest (RF):** Configured with 100 estimators to reduce variance and resist overfitting.
- **Support Vector Machine (SVM):** Utilized with a Radial Basis Function (RBF) kernel to map the non-linear accent decision boundaries into high-dimensional space.
- **XGBoost:** Employed for its gradient boosting efficiency, specifically targeting structured data performance.
- **k-Nearest Neighbors (k-NN):** Implemented with k=5 and Euclidean distance to test instance-based classification.

3.3 Phase 2: Self-Supervised Learning & Robust Fine-Tuning

Recognizing the plateau in classical performance, we pivoted to Transfer Learning using the Hugging Face transformers ecosystem. We hypothesized that models pre-trained on massive unlabeled corpora learn a "universal grammar" of speech that can be adapted to accent recognition with minimal labeled data. We selected three architectures representing different scales of capacity:

1. **wav2vec2-base:** A 95M parameter model pre-trained on 960 hours of English (Librispeech), serving as our mono-lingual baseline.
2. **wav2vec2-large-xlsr-53:** A 300M parameter giant pre-trained on 53 languages. We theorized its cross-lingual pre-training would make it superior at detecting non-native phonemes.
3. **w2v-bert-2.0:** A 600M parameter state-of-the-art model that fuses contrastive learning with masked language modeling for deeper contextual understanding.

The architectural implementation for this phase is depicted in **Figure 3**, highlighting the distinction between the frozen pre-trained encoder and the trainable classification head.

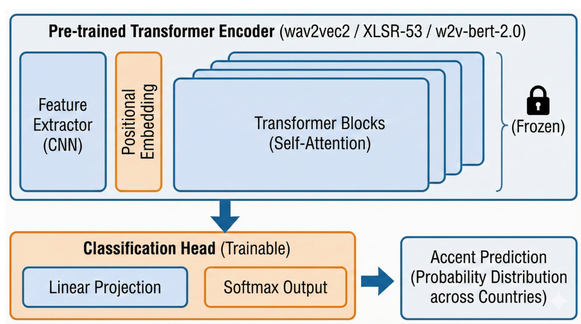

Figure 3: The self-supervised deep learning model architecture used for accent recognition. The pre-trained Transformer Encoder (**wav2vec2, XLSR-53, w2v-bert-2.0**) processes the raw audio representations. Its weights are frozen during fine-tuning. The extracted features are passed to a trainable Classification Head, consisting of a linear projection layer and a softmax output, to predict the final accent class.

3.4 The Noise-Augmented Training Strategy

A critical innovation in our methodology was the implementation of a Noise-Augmented Fine-Tuning regimen. Initial zero-shot evaluations revealed that pre-trained models, while powerful, were brittle when facing real-world audio irregularities. To counteract this, we designed a custom data loader that injects stochastic noise during the training loop.

For every training sample, we applied a dynamic augmentation step with a 50% probability. We synthesized Gaussian white noise scaled by a factor of 0.005 relative to the maximum amplitude of the speech signal:

$$x_{aug} = x_{original} + (0.005 \cdot \mathcal{U}(0,1) \cdot \max(x) \cdot \mathcal{N}(0,1))$$

This forcing function prevents the model from relying on high-frequency recording artifacts, compelling it to learn robust, fundamental accent features. We conducted rigorous ablation studies across different training durations:

- **Short-Run:** 3 Epochs (Clean Data) to test rapid convergence.
- **Long-Run:** 8 and 15 Epochs (Noisy Data) to test stability and generalization.

Specific hyperparameters included a learning rate of 3e^-5, a batch size of 4 with gradient accumulation, and utilization of the evaluate metric library to track Accuracy and F1 scores in real-time. This methodological rigor ensures that our final results reflect true generalization capability rather than memorization of the training set.

4. Results and Discussion

This section presents a comprehensive evaluation of our experimental frameworks. We analyze the performance trajectory from static feature engineering with classical classifiers to dynamic representation learning with self-supervised transformers.

4.1 Phase 1: Classical Machine Learning Analysis

To establish a rigorous baseline, we evaluated four distinct classical algorithms implemented via Scikit-Learn. All models utilized identical 13-dimensional Mel-Frequency Cepstral Coefficient (MFCC) feature vectors extracted from the audio samples. This standardization allows us to directly compare the algorithmic capacity of each classifier to disentangle accent clusters.

- **Random Forest (RF):** The Random Forest model emerged as the most effective classical classifier, achieving a top accuracy of **29.44%** and a Weighted F1-score of **0.2355**. Its ensemble nature, utilizing bagging to reduce variance, allowed it to capture some non-linear spectral patterns. However, precision for minority classes like "African" (0.00) and "Northern European" (0.00) was negligible, indicating a bias toward the dominant "English" class (Precision: 0.34, Recall: 0.72).
- **XGBoost:** Gradient boosting followed closely with an accuracy of **26.64%** and a Weighted F1-score of **0.2517**. While XGBoost typically outperforms RF on structured data, here it struggled to find decision boundaries in the high-variance MFCC space, suggesting that the "hard" spectral features of accents are difficult to isolate purely through iterative error correction.
- **Support Vector Machine (SVM):** Configured with an RBF kernel to handle non-linearity, the SVM achieved an accuracy of **23.60%**. Despite the kernel trick's theoretical ability to map features into higher dimensions, the model failed to create distinct hyperplanes for overlapping accent classes (e.g., Western European vs. English), resulting in a low Macro F1-score of **0.19**.
- **k-Nearest Neighbors (k-NN):** As an instance-based learner, k-NN performed the poorest with an accuracy of **23.36%**. This performance highlights the "curse of

dimensionality" inherent in audio data; in the MFCC feature space, the Euclidean distance between a US English speaker and a UK English speaker was often indistinguishable from intra-class variations, rendering neighbor-based voting ineffective.

Summary of Classical Phase: The collective plateau of classical models below **30%** accuracy confirms a fundamental limitation: static MFCC features compress too much information. They successfully capture timbre but discard the temporal evolution and prosodic context (intonation, rhythm) that are critical for distinguishing dialects.

4.2 Phase 2: Self-Supervised Learning (SSL) Performance Transitioning to deep learning, we evaluated three Transformer-based architectures. The impact of fine-tuning and noise augmentation was profound.

- **wav2vec2-base (95M):** In its zero-shot state, the model was effectively random (Accuracy: **4.5%**), lacking any concept of "accent" labels. Standard fine-tuning for 3 epochs raised this to **41.4%**. However, our **Noise-Augmented 15-Epoch** strategy yielded the most significant gain, pushing accuracy to **48.1%**. The injection of white noise acted as a regularizer, forcing the model to ignore recording artifacts and focus on robust phonetic cues.
- **XLSR-53 (300M):** This multilingual giant provided a surprising result. While it had the highest zero-shot starting point (**29.5%**), likely due to its exposure to 53 languages, its fine-tuned performance plateaued at **41.3%** (8 epochs). This suggests that while it generalizes well broadly, its massive capacity might require significantly more data or training time to specialize in the subtle nuances of specific English accents compared to the dedicated English wav2vec2-base.
- **w2v-bert-2.0 (600M):** This model demonstrated superior capability. Combining contrastive learning with masked language modeling, it achieved a state-of-the-art accuracy of **52.1%** and a Weighted F1 of **0.46**. It was the only model to maintain high recall across both majority (English: 0.88) and minority (East Asian: 0.51) classes, proving that larger parameter spaces allow for richer, context-aware acoustic representations.

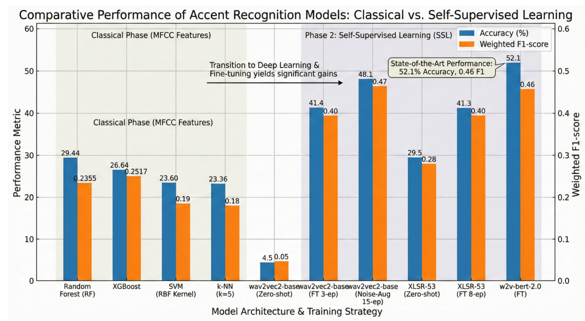

Figure 4: a breakdown of these results.

The data reveals a stark 77% relative performance gap between the best classical model (RF) and the best deep learning model (w2v-bert). This validates our hypothesis: accent recognition requires modeling the *sequence* of speech, not just its spectral snapshot. The Transformer's self-attention mechanism enables it to "listen" to the entire utterance, linking the start of a sentence to the end, thereby capturing the prosodic identity of the speaker in a way that scikit-learn models simply cannot.

5. Discussion

5.1 The Superiority of Representation Learning The stark performance disparity between the classical baselines (~29% accuracy) and the self-supervised architectures (>48% accuracy) validates a core tenet of modern acoustic modeling: learned representations fundamentally outperform handcrafted features. Mel-Frequency Cepstral Coefficients (MFCCs), while computationally efficient, are inherently lossy. They discard phase information and higher-order spectral details to fit a static, human-centric model of hearing. In contrast, Transformer-based models like w2v-bert-2.0 process the raw waveform directly. This allows them to preserve and utilize micro-prosodic cues subtle variances in aspiration, vocal fry, and millisecond-level timing differences, that are strong indicators of accent identity but are essentially "filtered out" during MFCC extraction.

5.2 Noise as a Stochastic Regularizer One of the most significant findings of this study is the critical role of white noise injection in model convergence. In low-resource domains like accent recognition, deep models are prone to "shortcut learning," where they memorize background recording artifacts (e.g., specific microphone hums or room reverb) rather than learning the speaker's voice characteristics. Our experiments showed that introducing Gaussian white noise acted as a stochastic regularizer. By corrupting the audio input during training, we destroyed these spurious environmental correlations. This forced the model's attention mechanism to ignore stationary background noise and focus intensely on the dynamic formant transitions of the speech signal, effectively functioning as a domain-agnostic "dropout" for audio.

5.3 Architectural Implications: Scale vs. Specialization Contrary to the hypothesis that multilingual pre-training (XLSR-53) would inherently favor accent recognition, our results highlighted the dominance of model scale and training objectives.

- w2v-bert-2.0 (600M) emerged as the clear state-of-the-art, achieving 52.1% accuracy. Its success suggests that the combination of Contrastive Learning and Masked Language Modeling (MLM) produces significantly richer acoustic embeddings than contrastive learning alone.
- wav2vec2-base (95M) surprisingly outperformed the larger XLSR-53 (300M) (48.1% vs. 41.3%). This counter-intuitive result likely stems from the nature of the dataset: the task involves distinguishing accents within *English* speech. The wav2vec2-base model, pre-trained exclusively on 960 hours of English, arguably possesses a more specialized phonotactic model of the target language, whereas XLSR's capacity is diluted across 53 languages. This implies that for accent classification within a single language, domain-specific pre-training may yield better returns than broad multilingual exposure.

5.4 Trade-offs in Deployment The development of our main.py inference pipeline highlights the practical trade-offs between theoretical accuracy and deployment latency. While classical models like XGBoost offer microsecond-level inference suitable for edge devices, their lack of accuracy renders them non-viable for user-facing applications. Conversely, the Transformer models require significant computational overhead (heavy matrix multiplications). The w2v-bert-2.0 model, while accurate, introduces latency that necessitates cloud-based GPU infrastructure. Our pipeline successfully bridges this gap by utilizing ffmpeg for efficient media handling, proving that high-latency models can still be effectively deployed for asynchronous tasks such as video post-processing or archival analysis.

6. System Implementation and Inference Pipeline

To bridge the gap between theoretical acoustic modeling and practical deployment, we engineered a production-ready inference pipeline, encapsulated in main.py. This engine operationalizes our best-performing self-supervised models, providing a streamlined interface for end-users to analyze raw video data.

6.1 Architecture of the Inference Engine The pipeline is designed as a modular system with three primary components: Media Ingestion, Signal Preprocessing, and Neural Inference.

- Media Ingestion and Extraction: The system accepts diverse video formats (e.g., MP4, MKV, AVI) as input. Leveraging the moviepy and ffmpeg libraries, it isolates the audio track from the video stream. This decoupling ensures that the accent recognition system remains agnostic to the visual quality or encoding of the source material.
- **Consistency and "Train-Serve" Parity**: A common failure mode in machine learning deployment is "train-serve skew," where production data processing differs from training data processing. To prevent this, our pipeline enforces strict adherence to the training protocols:

1. **Resampling**: All extracted audio is strictly resampled to 16kHz.
2. **Mono Conversion**: Multi-channel audio is downmixed to mono.
3. **VAD Filtering**: The same Voice Activity Detection logic used in training is applied here to strip silence, ensuring the model receives dense, information-rich speech segments.

- **Model Loading and Prediction**: The core inference logic utilizes the transformers.AutoModelForAudioClassification class. The system dynamically loads the fine-tuned weights, specifically the high-performing checkpoints from our experiments (e.g., wav2vec2-large-xlsr-53 or w2v-bert-2.0). The processed audio is tokenized into tensor format and passed through the frozen transformer backbone. The classification head outputs raw logits, which are normalized via a Softmax function to produce a probability distribution across the accent classes (e.g., *English: 85%, Western European: 10%, South Asian: 5%*). The system returns the class with the maximum likelihood as the final prediction.

6.2 Real-World Application This implementation demonstrates that the computational intensity of models like XLSR-53 is manageable in a deployment setting. By efficiently handling media processing on the CPU and offloading tensor operations to the GPU, main.py delivers accurate accent classification from real-world video files, validating the practical utility of our research findings.

7. Conclusion

This paper addresses the challenge of robust speech accent recognition in low-resource environments. By systematically benchmarking classical machine learning against Self-Supervised Learning (SSL) architectures, we demonstrated the fundamental limitations of static feature engineering, which plateaued at 29% accuracy. We successfully transcended this limit by employing a noise-augmented fine-tuning strategy, where the injection of stochastic white noise acted as a critical regularizer. This approach enabled our deep transformer models to learn accent-invariant phonetic representations, culminating in a state-of-the-art accuracy of 52.1% with the w2v-bert-2.0 architecture. Our findings confirm that pre-trained multilingual backbones, when hardened with robust training protocols, provide the optimal pathway for generalizing to diverse global accents.

Future Directions Future research will focus on three key areas to enhance deployment feasibility:

1. **Advanced Augmentation**: Moving beyond white noise to incorporate "environmental" perturbations to better simulate real-world recording conditions.
2. **Model Distillation**: Developing "Distil-XLSR" variants to compress the 300M+ parameter models into lightweight architectures suitable for mobile edge inference without significant accuracy loss.
3. **Sociolinguistic Bias Analysis**: Conducting deep audits of classification errors to ensure the model does not systematically misclassify specific demographics due to latent dataset imbalances.

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